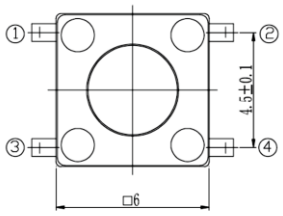
ECE 2305 – Introduction to C Programming
Hardware Project 02
7-Segment Display

Project Features: Digital Input and Digital Output.

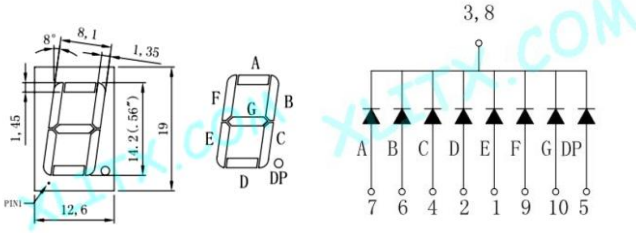
Program an Arduino board to accept a 4-bit binary input and display the numerical value (0 to 9) on a 7-Segment LED display.

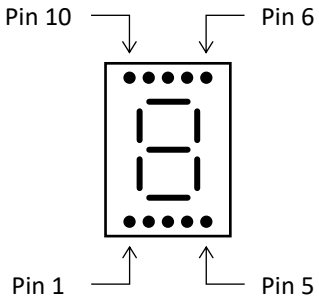
Use the pushbutton for the digital input:



ACTUATOR POSITION ( ): MOMENTARY	
NORMAL	DOWN
OFF	(ON)
SCHEMATIC	

Use the 7-Segment LED Display:



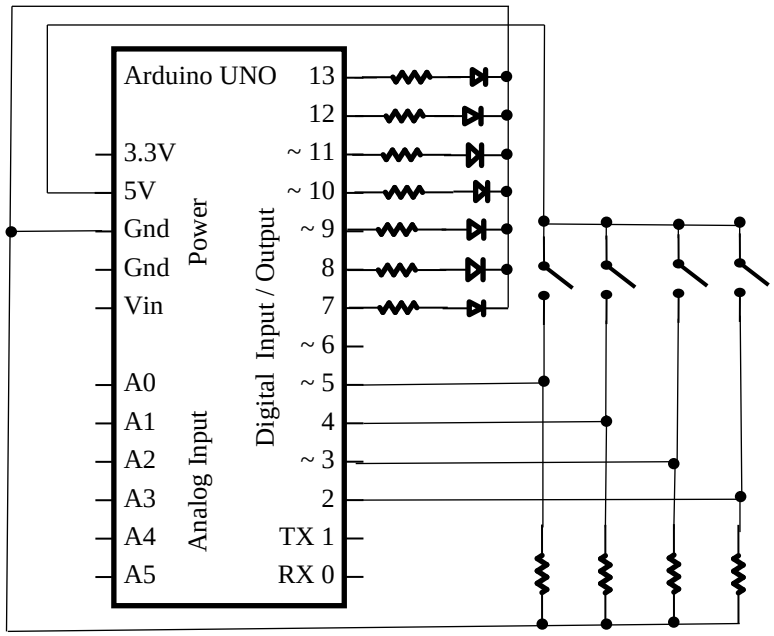


To document your program, create a document and include the following sections in the document. Provide a brief description of the system and how you are designing it to operate,

ECE 2305 – Introduction to C Programming
Hardware Project 02
7-Segment Display

A. Hardware Diagram:

Provide a hardware diagram of the components.

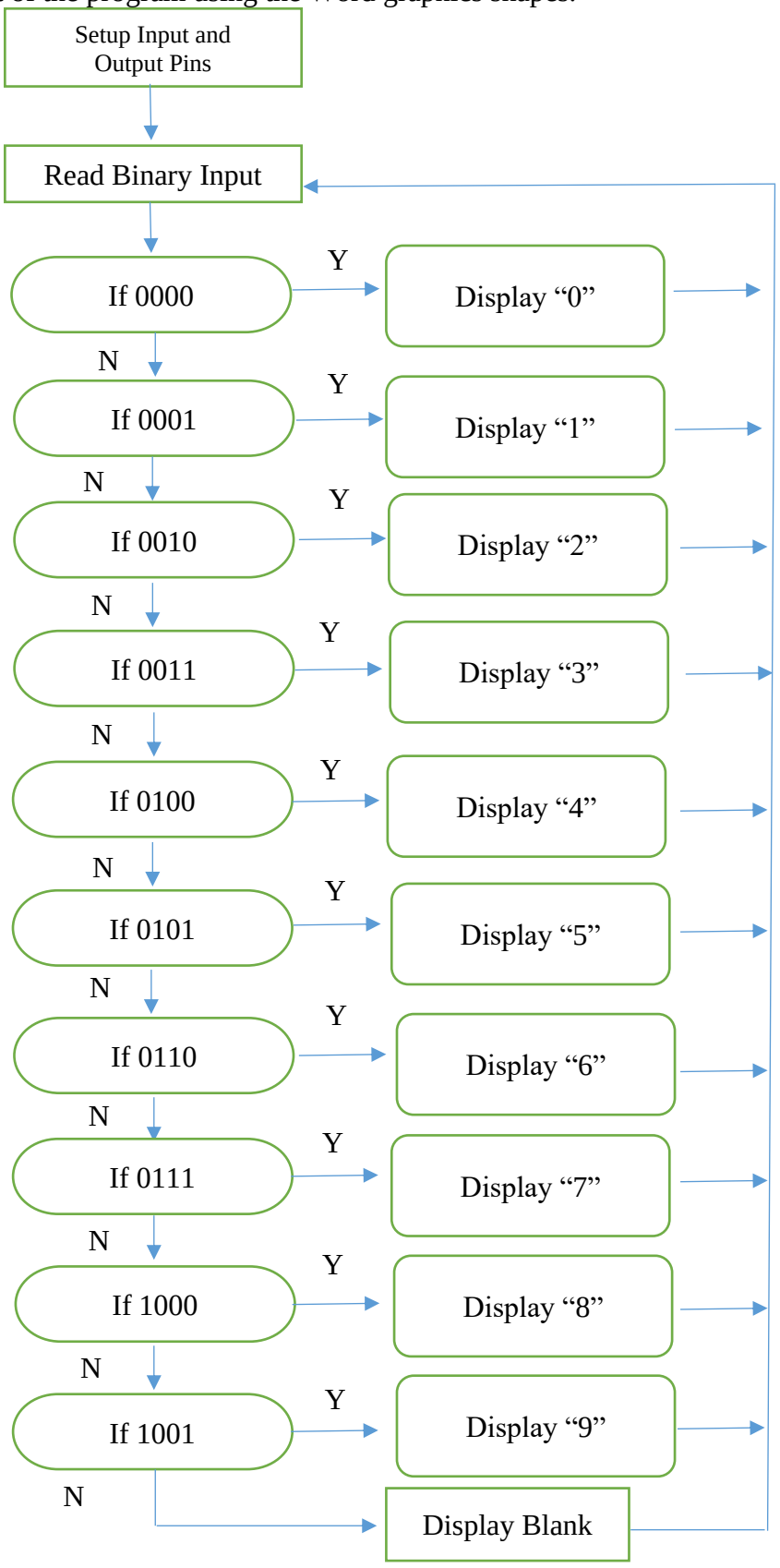


Hardware Diagram

ECE 2305 – Introduction to C Programming
Hardware Project 02
7-Segment Display

B. Program Flowchart:

Draw a flowchart of the program using the Word graphics shapes.



C. Arduino Source Code

```
// Hardware Project 2 - 7-Segment Display with 4-bit Binary Input

void setup()
{
  // Setup 7-segment display output pins
  pinMode(13, OUTPUT); // segment g
  pinMode(12, OUTPUT); // segment f
  pinMode(11, OUTPUT); // segment e
  pinMode(10, OUTPUT); // segment d
  pinMode(9, OUTPUT);  // segment c
  pinMode(8, OUTPUT);  // segment b
  pinMode(7, OUTPUT);  // segment a

  // Setup binary input pins
  pinMode(5, INPUT); // bit 3
  pinMode(4, INPUT); // bit 2
  pinMode(3, INPUT); // bit 1
  pinMode(2, INPUT); // bit 0
}

void loop()
{
  // Read the 4-bit binary input and convert to decimal
  int number = (digitalRead(5) << 3) | (digitalRead(4) << 2) | (digitalRead(3) <<
1) | digitalRead(2);

  // Display the corresponding number on the 7-segment display
  if (number == 0)
  { digitalWrite(13, LOW); digitalWrite(12, HIGH); digitalWrite(11, HIGH);
digitalWrite(10, HIGH);
    digitalWrite(9, HIGH); digitalWrite(8, HIGH); digitalWrite(7, HIGH); }

  else if (number == 1)
  { digitalWrite(13, LOW); digitalWrite(12, LOW); digitalWrite(11, LOW);
digitalWrite(10, LOW);
    digitalWrite(9, HIGH); digitalWrite(8, HIGH); digitalWrite(7, LOW); }

  else if (number == 2)
  { digitalWrite(13, HIGH); digitalWrite(12, LOW); digitalWrite(11, HIGH);
digitalWrite(10, HIGH);
    digitalWrite(9, LOW); digitalWrite(8, HIGH); digitalWrite(7, HIGH); }

  else if (number == 3)
```

```
  { digitalWrite(13, HIGH); digitalWrite(12, LOW); digitalWrite(11, LOW);
digitalWrite(10, HIGH);
    digitalWrite(9, HIGH); digitalWrite(8, HIGH); digitalWrite(7, HIGH); }

  else if (number == 4)
  { digitalWrite(13, HIGH); digitalWrite(12, HIGH); digitalWrite(11, LOW);
digitalWrite(10, LOW);
    digitalWrite(9, HIGH); digitalWrite(8, HIGH); digitalWrite(7, LOW); }

  else if (number == 5)
  { digitalWrite(13, HIGH); digitalWrite(12, HIGH); digitalWrite(11, LOW);
digitalWrite(10, HIGH);
    digitalWrite(9, HIGH); digitalWrite(8, LOW); digitalWrite(7, HIGH); }

  else if (number == 6)
  { digitalWrite(13, HIGH); digitalWrite(12, HIGH); digitalWrite(11, HIGH);
digitalWrite(10, HIGH);
    digitalWrite(9, HIGH); digitalWrite(8, LOW); digitalWrite(7, HIGH); }

  else if (number == 7)
  { digitalWrite(13, LOW); digitalWrite(12, LOW); digitalWrite(11, LOW);
digitalWrite(10, LOW);
    digitalWrite(9, HIGH); digitalWrite(8, HIGH); digitalWrite(7, HIGH); }

  else if (number == 8)
  { digitalWrite(13, HIGH); digitalWrite(12, HIGH); digitalWrite(11, HIGH);
digitalWrite(10, HIGH);
    digitalWrite(9, HIGH); digitalWrite(8, HIGH); digitalWrite(7, HIGH); }

  else if (number == 9)
  { digitalWrite(13, HIGH); digitalWrite(12, HIGH); digitalWrite(11, LOW);
digitalWrite(10, HIGH);
    digitalWrite(9, HIGH); digitalWrite(8, HIGH); digitalWrite(7, HIGH); }

  else
  { digitalWrite(13, LOW); digitalWrite(12, LOW); digitalWrite(11, LOW);
digitalWrite(10, LOW);
    digitalWrite(9, LOW); digitalWrite(8, LOW); digitalWrite(7, LOW); }
}
```

D. Demonstration Video

Record and upload a video demonstrating the operation of the circuit. (DONE)

Save the document as a *PDF* file and submit the *PDF* document to *Blackboard*.